PYTHON PROGRAMMING
ALA-2

TOPIC: LIST SLICING:

Interview style Q & A.

>>>> 1. What is list slicing in Python?

Answer:

List slicing is a way to extract a portion of a list using the syntax:
list[start:stop:step]

>>>> 2. What does a[1:4] return?

Answer:

It returns elements from index 1 to 3 (stop index excluded).

```
a = [10, 20, 30, 40, 50]
print(a[1:4])  # [20, 30, 40]
```

>>>> 3. What happens if start index is omitted?

Answer:

Python assumes start = 0.

```
a[:3]  # first 3 elements
```

>>>> 4. What happens if stop index is omitted?

Answer:

Python slices till the end of the list.

```
a[2:]  # from index 2 to end
```

>>>> 5. What does a[:] do?

Answer:

It creates a shallow copy of the list.

>>>> 6. What is the use of step in slicing?

Answer:

Step defines how many elements to skip.

```
a[::2]  # every second element
```

>>> 7. What does negative slicing mean?

Answer:

It accesses elements from the end of the list.

a[-1] # last element

>>> 8. What does a[::-1] do?

Answer:

It reverses the list.

>>> 9. What is the output of a[3:1]?

Answer:

It returns an empty list because slicing moves forward by default.

[]

>>> 10. How to slice a list in reverse order using step?

Answer:

Use a negative step.

a[5:1:-1]

>>> 11. What is the difference between a[:] and a?

Answer:

- a[:] creates a new copy
- a refers to the same list

>>> 12. Can slicing go out of index range?

Answer:

Yes, Python handles it safely without error.

a[0:100] # no error

>>> 13. How to replace elements using slicing?

Answer:

You can assign new values to a slice.

a[1:3] = [99, 100]

>>> 14. What happens when step is 0?

Answer:

It raises an error:

ValueError: slice step cannot be zero

>>> 15. How to extract alternate elements starting from index 1?

Answer:

a[1::2]